Market Split based Congestion Management for Networks with Loops

Marta Marmiroli Member (Mitsubishi Electric Corporation)
Masahiko Tanimoto Member (Mitsubishi Electric Corporation)
Yukitoki Tsukamoto Member (Mitsubishi Electric Corporation)
Ryuichi Yokoyama Member (Waseda University)

Keywords: electricity market, transmission network, congestion, loops, single price auction, market splitting

The problem of transmission congestion is one of the peculiarities of the power industry. Due to deregulation of the market and the increase of the number of transmission users, the utilization of the interconnections has changed creating congestion. Market players dispatch generating resources with the objective to increment their own economic benefit. For this reason, it is common that the transfer capacity required to transfer electricity from a control area to another control area exceeds the transfer capability of the interconnection line between the two areas.

In case of insufficient transfer capacity among control areas, it is then necessary to introduce rules to manage the congestion on the transmission lines. An effective approach to this problem for auction-based markets is the market split approach.

The market split is recognized as a simple method able to relief congestion in a fair manner.

To solve the market split model for power system network with radial interconnections and loops, the authors developed an algebraic algorithm that ensure the optimal utilization of the available transmission capacity among areas. The algorithm is based on the definition of three types of congested areas: area congested in export (E), area congested in import (I) and area non-congested or neutral (N). In Fig. 1 all the combination of adjacent areas is described.

The developed method is utilized for day ahead spot market. To clear the market a single-price auction algorithm is implemented. The model fit properly with the electricity spot market because of its simplicity and the possibility to treat two sides auction (supply and demand). The single price auction was also pointed by FERC as the only method to be utilized under the SMD (Standard Market Design) for spot market.

To find the solution that optimize the transmission utilization and maximize the benefit of the market participants, an iterative method is used. First the market is cleared without transmission constraints: orders are aggregated in a sell curve and buy curve and the market clearing price is determined as the intersection of the two curves; in

From \ To	Import Area	Export Area	Neutral Area
Import Area	I → I	I → E	I → N
Export Area	E → I	E → E	E → N
Neutral Area	N → I	N → E	N → N

Fig. 1. Areas combination

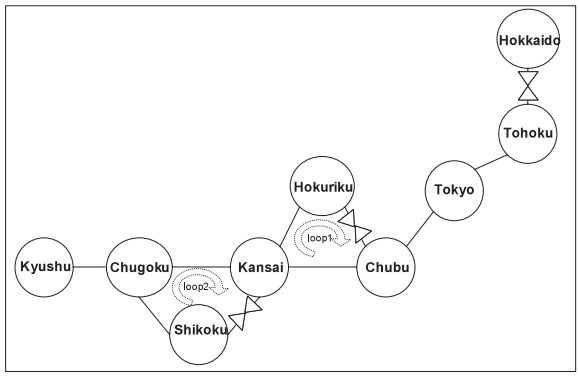


Fig. 2. Representation of the Japanese network

the next step the power flow for each transmission line is calculated with a simple power flow calculation. Based on the transmission constraints and the power flow data, the type of each area is determined. In case of congestion between an export area and an import area with power flow from the export area to the import area, the market is divided in 2 or more submarkets. To ensure the optimal utilization of the transmission capacity, a virtual order with the same value of the transmission capacity is introduced in each submarket (sell order in the import submarket, buy order in the export submarket).

For each submarket, the process is repeated until all the transmission congestions are relieved.

Under the assumption that at least on interconnection in a loop is a DC interconnection, networks with loops are transformed in an equivalent network with a delta-star transformation. The transformation is useful to model the Japanese network where two loops are present as described in Figure 2.

A virtual area is introduced in the center of the star and treated as any other area already defined in the system.

It is generally recognized that the day-ahead market is operated under a tight schedule, for this reason it is fundamental to get the market clearing solution in a very short time period. The objective to develop this new algorithm, as substitution of more common linear programming approach, is to improve the time calculation for markets with a large number of participants and several areas.

The algorithm was tested on large-scale problems with more than 80,000 of market orders (bids and offers), 16 areas and 15 transmission interconnections. The solution was produced in 0.01 second on an Itanium 1 GHz CPU confirming the high performance of the algorithm.

Paper

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Marta Marmiroli* Member
Masahiko Tanimoto** Member
Yukitoki Tsukamoto* Member
Ryuichi Yokoyama*** Member

Market splitting is one of the methods to solve the transmission congestion problem associated with the introduction of competitive electricity market and transmission access. Based on the concept of price difference among congested areas, the market splitting approach produces a solution that strongly informs market participants of congestion path. In this paper, an algorithm to solve the market splitting problem for complex networks including loop structures is proposed. The method, based on an algebraic approach, ensures a feasible optimal solution verifiable and easily understandable by the market participants. Complex networks are transformed into simple radial ones using the delta-star approach. The method was tested on large problems to evaluate the performances.

Keywords: electricity market, transmission network, congestion, loops, single price auction, market splitting

1. Introduction

The rapid deregulation of the electricity industry in the last two decades has produced the formation of several markets around the world. Following the example of other commodities, the electricity market is commonly managed by an exchange, which publishes the traded quantities and the market prices on a daily base. However, electricity cannot be exchanged freely because it has to be traded subject to the physical constraints of the transmission network. These constraints can cause congestions.

The problem of transmission congestion in Japan is especially relevant to interconnections among power system control areas.

The interconnected power system networks are designed to meet the operational requirements such as frequency control, stability support, and emergency procedure. Therefore, the capacity operation of the interconnection is originally designed not for competitive trading but for the previously described security operation. However, due to deregulation of the electricity industry and the increase of the number of transmission users, the utilization of the interconnections has changed: not only the security control but also trading support. Market players dispatch generating units with the objective to maximize their own economic benefit. For this reason, it is common that the transfer capacity required to transfer electricity from a control area to another control area exceeds the transfer capability of the interconnection line between the two areas.

In other words, it can be said that the transmission congestion problem is caused by the conflict of interests among the market players: limited transmission resources have to be allocated to several independent players, all of them operating with the intent to improve their own profit.

In case of insufficient transfer capacity among control areas, it is then necessary to introduce rules to manage the congestion on the transmission lines. A congestion management method has to be able to relief the overflow on the transmission lines and, at the same time, to give a strong economical signal to the transmission users. This economic signal is worldwide recognized as the difference of the electricity price between congested areas.

Several market structures such as Nord Pool in north Europe, EEX in Germany and Japanese PX [1] adopted a market split approach to manage congestion in the spot market. When overflow on an interconnection line between two control areas is identified, the spot market is divided into two separated independent sub-markets and a different price is assigned to each of them.

The market split model was approached with several algorithms and methods. The most common method is to formalize the problem as a linear maximization one and to solve it with a linear programming approach [2][3]. In addition non-linear formulation [4] and iteratively algebraic approach [5] were also proposed. These methods produces all the same solution but they can be time consuming and it is difficult to explain their process to draw the solution to market participants.

In this paper, a new algebraic algorithm to solve the market split problem is proposed; the algorithm may be easily applied to practical interconnection networks including radial and loop structures.

This paper is structured as follow. In section 2, an introduction of the auction rules for electricity spot market is discussed, and the market split concept is described in the following section. The proposed algorithm to identify the

* System Integration Department, Mitsubishi Electric Corporation
2-7-3, Marunouchi, Chiyoda-ku, Tokyo 100-8310
** Advanced Technology R&D Center, Mitsubishi Electric Corporation
8-1-1, Tsukaguchi-Honmachi, Amagasaki 661-8661
*** Graduate School of Environment Energy Engineering, Waseda University
1-1-7, Nishi Waseda, Shinjuku-ku, Tokyo 160-0051

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congested sub-market is described in section 4, and a numerical example is illustrated in section 5. After performance evaluation, conclusions and open problems are discussed in section 7.

2. Auction Rules

In this paper, a single price auction is introduced as mechanism to clear the day-ahead spot market. This auction, which is known as uniform price auction, was originally proposed by Friedman [6] to solve the problem of allocation of multiple units. The model properly fits with the electricity spot market because of its simplicity and the possibility to manage double sides bidding. The single price auction was also pointed by Federal Energy Regulatory Commission (FERC) as the only method to be applied for spot market under the Standard Market Design [7].

The objective of the single price auction is to find the market price that maximizes the exchange volume in the electricity market and to allocate the individual quantity that may be contracted by each player. The price is discovered at the cross point between the supply and demand curves, and all the electricity exchanged is settled at the same price.

An order for the market is defined with the pair of price-quantity and its position. An offer order is a will to sell a certain quantity of electricity at a price equal or higher than the offer price, and oppositely a bid order is a will to buy a certain amount of electricity at a price equal or lower than the bid price. Participants' orders are collected until a predetermined time, and the orders are aggregated to build the aggregated supply and demand curves. The market price is determined in the cross point between the aggregated supply and demand curves and the contract quantity for each order are allocated. However, the maximization process of the exchange quantity does not ensure the uniqueness of the solution, and it may happen that the total exchange quantity is maximum in an interval of prices. An additional rule, therefore, is necessary to fix the unique solution. It is assumed that in case of multiple solutions, the lowest price that satisfies the previous described constraints is selected as market price. In some cases, the total offer quantity may not be equal to the total bid quantity at the market price. The excess of quantity, in the offer or in the bid side, is defined as a surplus. The surplus is distributed among order with price equal to the market price in a proportional way.

The authors implemented a sequential approach to determine the market price and allocate the contract quantity for each order. The sequential method is described in the following steps:

(1) sort all the bid and offer prices from the highest to the lowest.
(2) calculate the aggregated sell and buy quantity for each price.
(3) calculate the tradable quantity for each price as the minimum between the aggregated sell quantity and buy quantity
(4) select the price at which correspond the maximum tradable quantity. If two or more prices have the same tradable quantity, the price with the smallest surplus (aggregated sell quantity minus aggregated buy quantity) is selected.

Once the price and the total trade are individuated, the contract quantity for each player is calculated using the rules previous described: seller with a bid price lower than the market price will sell all the bid quantity; in a similar way, buyer with a bid price higher than the market price will buy all the bid quantity, seller and buyer with a bid price equal to the market price will contract a quantity between 0 and the bid quantity depending on the market availability. The market price, solution of the described algorithm, satisfies the condition to be the cross point between bids curve and offers curve.

The proposed method is a practical algorithm that can be easily implemented on a computer system. Similar algorithms are implemented and performed daily in commodities and stocks exchanges [8].

3. Market Splitting

One of the main differences between electricity and the other commodities is that electricity has to be transferred through transmission lines. The previously described auction algorithm does not consider any constraints related with the transmission network. The location information is another important factor for the order in addition to the quantity and the price in the physical electricity market. From a power system point of view, the location of an order can be considered as:

- The source point in case of a sell order
- The sink point in case of a buy order.

It may happen that the solution developed by the previously described algorithm may not respect any of the constraints on the transmission lines. In this case, the market should be split into two sub-markets with different prices, binding of transmission constraint in an optimal manner.

Figure 1 represents a simple example with 2 areas connected by a transmission line. A player in Area 1 offers a quantity of electricity equal to Q_{offer} at price P_{offer}, and a player in Area 2 bids a quantity of electricity equal to Q_{bid} at price P_{bid}. The trade quantity is described in the following formula, which corresponds to the power flowing between Area 1 and 2.

$$Q_{trade} = Q_{flow} = \begin{cases} \min(Q_{bid}, Q_{offer}) & \text{if } P_{bid} > P_{offer} \\ 0 & \text{otherwise} \end{cases} \quad \cdots\cdots (1)$$

If the trade quantity Q_{flow} is bigger than the transmission capacity Q_{limit}, the solution is not acceptable. In this case, the market has to be split into two sub-markets and the auction algorithm is executed for each of them once again. As a consequence, the market price in the exporting area should be lower or equal to the price without any consideration of transmission constraint. In opposite, the price in the importing

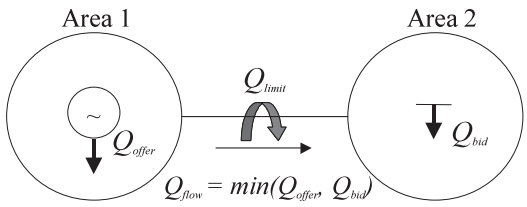


Fig. 1. Congested market example

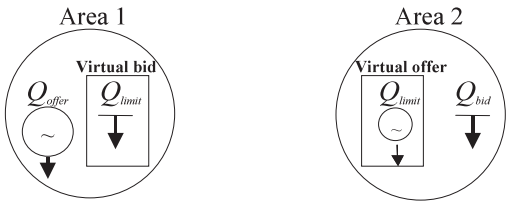


Fig. 2. Market split

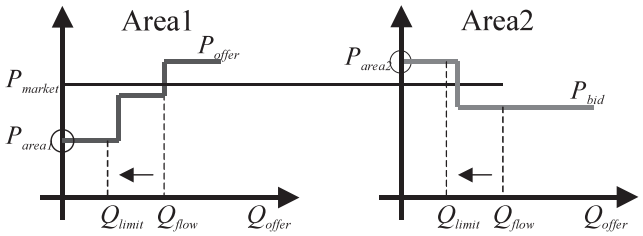


Fig. 3. Prices in the areas

area should be higher or equal to the unconstrained market price.

To execute the market split with the optimal transmission usage, the concept of virtual offer and bid is introduced.

A virtual offer is defined as the offer at the lowest price in the importing area with and the quantity Q_{limit}. A virtual bid is a bid at highest price in the importing area with the quantity Q_{limit}. Figure 2 shows how the market may be split with introducing virtual bid and offer.

The auction algorithm for Area 1 and 2 can be executed independently, which discover the market price and the total tradable quantity for each of them.

As represented in Fig. 3, since Q_{limit} is smaller than Q_{flow}, the price P_{area1} in Area 1 becomes lower than the market price P_{market} which is the price before the market splitting. On opposite, the price P_{area2} in Area 2 becomes higher.

Sequentially, the potential tradable quantity for each order is calculated, and the difference of net value of import and export should be calculated for each area:

$$p_i = \sum_{sell} Q_{sell}^{satisfied} - \sum_{buy} Q_{buy}^{satisfied} \quad \cdots\cdots\cdots\cdots\cdots\cdots (2)$$

where $Q_{sell}^{satisfied}$ is the total quantity of matched offers and $Q_{buy}^{satisfied}$ is the total quantity of matched bids in the area.

In case of radial networks, the flow on each transmission line can be easily calculated using the topology information and the quantities p_i.

In this paper, upper limits transmission constrains only are considered. In case of lower limits constraints (minimum flow for a transmission line), it is necessary to model the constrain as a band-constrain: first band between upper and lower limit, second band upper and lower limit equal zero. The authors intend to investigate this problem in the future.

4. Congested Sub-market Identification

In the previous section, the market split is applied for a market composed by two areas. In this case, it is easy to define the splitting point of the congested path. The problem becomes more complex in case of three or more areas. Figure 4 is an example of market with four areas. It is not trivial to define the location of the market split which maximizes the

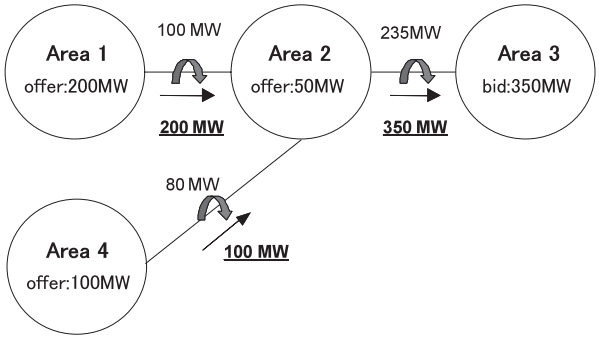


Fig. 4. Market with 4 areas and 3 congested lines

tradable quantity under the rules in section II.

It seems that the path between Area 2 and 3 is the most congested interconnection with an overflow of 115 MW. In reality, solving the congestion on path 1-2 and 4-2 will simultaneously relief the congestion on interconnection 2-3 without any needs to split the market between Area 2 and 3.

To individuate the position of the splits, a new algorithm is developed which is described in the next sections.

4.1 Area Type Definition First of all, in order to define the type for the area the concept of import, export, and neutral constrained are introduced. A parameter t_i for each area, which is named congestion index, is calculated as below,

$$t_i = p_i + \sum_j \min(Q_j^{limit}, Q_j^{flow}) - \sum_k \min(Q_k^{limit}, Q_k^{flow}) \quad \cdots\cdots\cdots\cdots\cdots (3)$$

where i is the index of the area, p_i is the net quantity of the area i, which is defined in Eq. (2), j is the index for interconnection with power flowing to the area i and k is the index for interconnection with power flowing from the area i, and further Q^{limit} is the maximum capacity and Q^{flow} is the quantity on the transmission line as result of the auction without any constraint on the network.

If the congestion index t is bigger than 0, an area is defined congested in export (E), and if t is lower than 0, congested in import (I), and further neutral (N) in case of t equal 0. From Eq. (3), it is understandable that the quantity of the importing and the net consumption in the area of congested in export is excess of the exporting quantity. In opposite, the quantity of the exporting and the net production in the area of the congested in import is less than the importing quantity.

An example of each kind of areas is described in Fig. 5. Due to the balance condition of sink and source of a power system, one or more areas should be an exporting congested one, if at least one import area exists and vice versa.

4.2 System Reduction and Splitting Rules In this section, the market splitting rules for multiple areas is described. Two areas with a transmission line should be split into two separated markets if and only if the area which is transferring electricity to the other area, is an export constrained area and the area which is receiving electricity, is an import area. The rule can be explained intuitively with the help of a graphical support in Fig. 6. All the possible combinations of areas are shown as nodes. An import node is represented with a barrier on the left that means the impossibility to get all the electricity required, an export node with

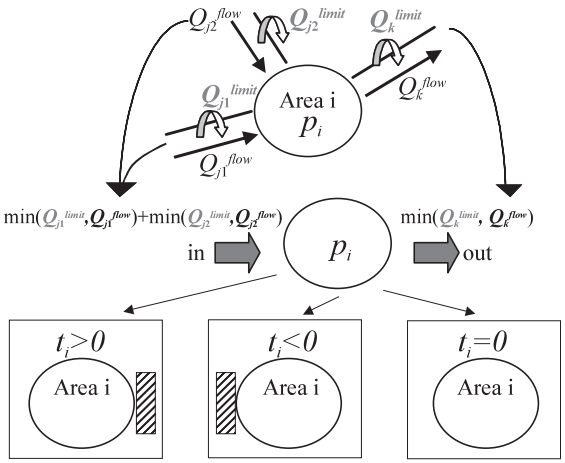


Fig. 5. Definition of export, import and neutral areas

From \ To	Import Area	Export Area	Neutral Area
Import Area	I → I	I → E	I → N
Export Area	E → I	E → E	E → N
Neutral Area	N → I	N → E	N → N

Fig. 6. The matrix of the splitting rules

a barrier on the right that means the impossibility to transfer all electricity that is required. A neutral node does not have any barrier. The arrow between the nodes represents the direction of the power flow on the line. From the figure, it can be said that in the case of Export-Import combination, the split of the two areas into two markets will has as consequence of a reduction of the power flowing between the two areas. In this case, the export node exports less power and the import area, imports less power reducing, in this way the congestion condition of both the two areas. On the contrary, in any other case in Fig. 6, the split of the two areas will not be of benefit for both or neither for one of the area (for example in the import-export combination). It is possible to demonstrate mathematical that splitting combination of import-export, import-import or export-export may cause an increase of system congestion and an inversion of prices (an area exporting electricity may have a market price higher than an area importing electricity) in case of radial network.

A system reduction algorithm is introduced in order to easily recognize the location of the export-import interconnection, which should be split. Two contiguous areas of the same kind are merged in one area. A neutral area has to be merged with an export area if it is contiguous to any, otherwise it can be merged with an import area. In Fig. 7, the reduction procedure is exampled. The connected areas 2 and 3, which are both import areas, are merged into a new area 2+3. After the merging, the system is reduced into a 3 areas with 2 interconnection of type export-import.

4.3 Loops Transformation The method, which is described in the previous section, is applicable only for radial network. In case of systems including the loop network, it is necessary to transform the topology in an equivalent radial network. For loop composed with 3 areas and at least one DC line, it is possible to apply a delta-star transformation

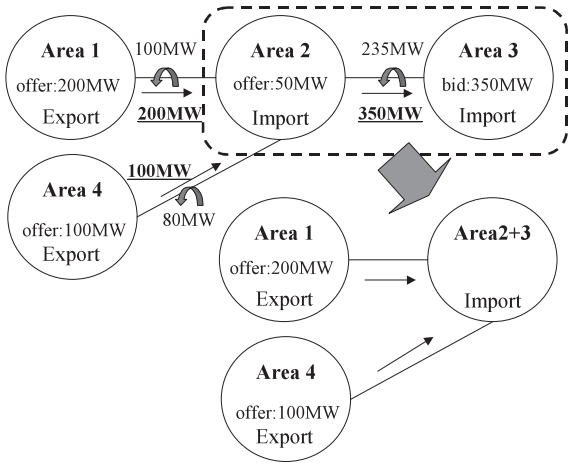


Fig. 7. Example of system reduction

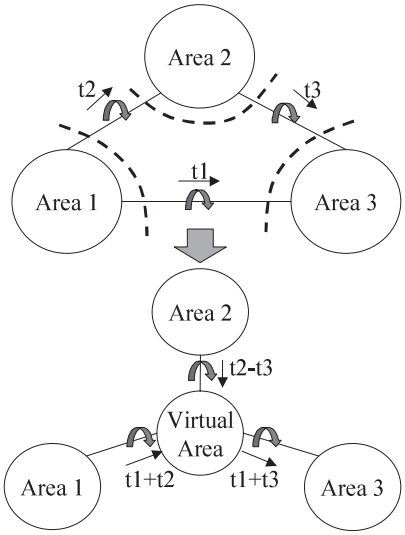


Fig. 8. Delta-Star transformation

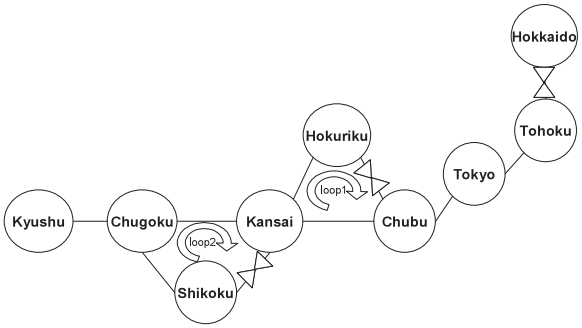


Fig. 9. Japanese network model for the spot market

with the introduction of a virtual, as described in Fig. 8. The capacity of each line in the star configuration is equal to the sum of capacity of the lines of the area itself. Similarly, the power flow can be calculated as the sum of the power flow on each line.

The virtual area is an area with net power production equal 0. Since it is assumed that at least on of the interconnection is a DC connection, the power flow of the loop can be solved without the introduction of the induction matrix but introducing a standard rule. In this paper, it is assumed the DC interconnection is utilized as much as possible.

In other world, the loop flow is calculated with the maximization of the power flow on the DC transmission line. The assumption to have only loops with at least one DC transmission is consistent with the Japanese network topology. The Japanese network model assumed by the spot market is described in Fig. 9.

Two loops are present in the network diagram but both of them have a DC interconnection.

5. Numerical Application

In this section, a market with 9 players on the system composed of 5 areas and 5 interconnections is solved; in Fig. 10, the system configuration and the player orders are shown. An order with a negative quantity is a sell order, on opposite a positive quantity is associated to a buy order. In area 1 and 2, relatively low price electricity is offered, and buy orders are mainly located in area 4 and 5. The interconnection between area 3 and 4 is a DC line.

In Table 1, the sorting procedure for the order based on the price and the price discovery of the market without any network constraints is presented. The tradable quantity reaches its maximum value at price of 8.5 and the total associated quantity is 650 MW.

The trading quantity for each player can be calculated with the price of 8.5. Players 1, 3, 5, and 8 offers are at lower price than the market price. Therefore, they are able to sell electricity. Further, because player 4 offered at 8.9, which is higher than the market price, his tradable quantity is 0.

The solution of the flow calculation with the market price 8.5 and the total tradable quantity of 650 MW is graphically described in Fig. 11. A large quantity of electricity is exported from area 1 and 2 into area 4 and 5. In addition of the power flow data and the tradable quantities for each player are also shown in Fig. 11.

The power flow on lines 2-3, 3-4, and 2-4 cannot be determinate as a unique solution. Applying the delta-star transformation, the system can be modeled as in Fig. 12.

Defining the areas type and in the process of the system reduction, it is evident that area 1, 2, 3, and the virtual area can be reduced into an export congested area and area 4 and 5 into an import congested area. The system can be then divided into two submarkets, the first composed of area 1, 2 and 3, the second of area 4 and 5 as described in Fig. 13. In order to divide the submarkets, it is necessary to introduce the virtual offers and bids. In area 2, a virtual bid of 150 MW and price of 9 is introduced, similarly in area 3 a virtual bid of 150 MW and the price of 9 is build. The price 9 is calculated adding 0.1 to the highest bid in submarket composed by area 1, 2 and 3. A virtual offer of 300 MW is introduced in area 4. The two submarkets can now be solved independently using the algorithm described in section 2.

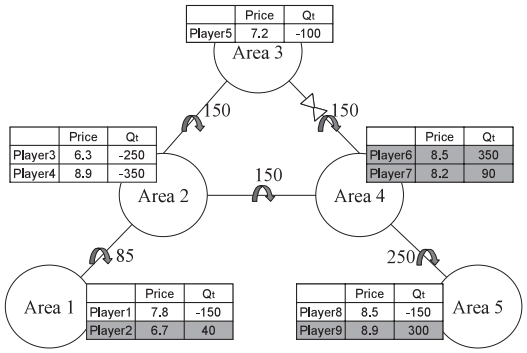


Fig. 10. Example system

Table 1. Market solution without network constraints

Aggregated sell	Sell	Price	Buy	Aggregated buy	Tradable quantity
1000	350	8.9	300	300	300
650	150	8.5	350	650	650
500		8.2	90	740	400
500		8		740	400
500	150	7.8		740	400
350	100	7.2		740	150
250		6.7	40	780	150
250	250	6.3		780	150

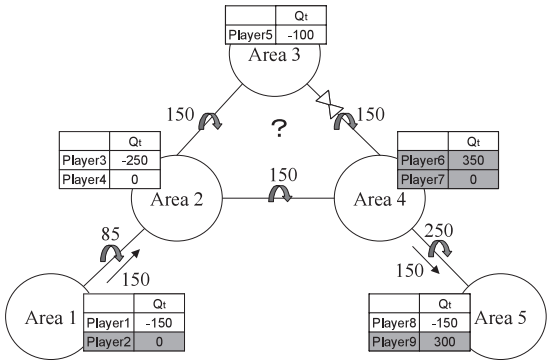


Fig. 11. Power flow

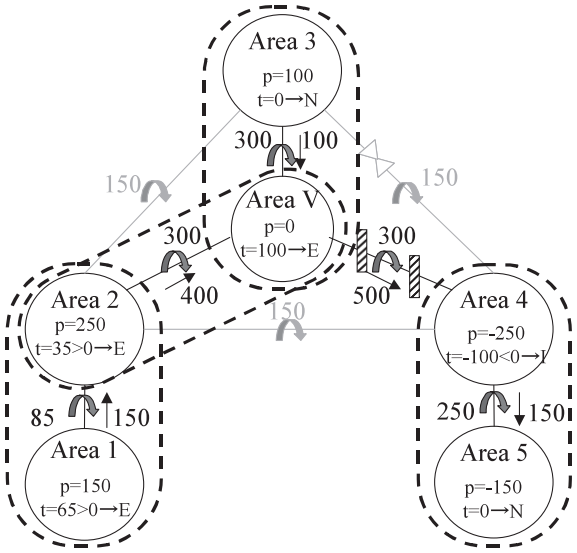


Fig. 12. Area types and system reduction

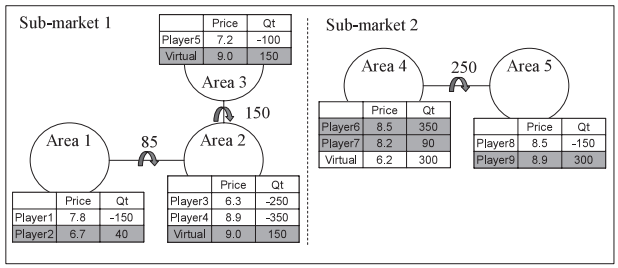


Fig. 13. Submarkets after splitting

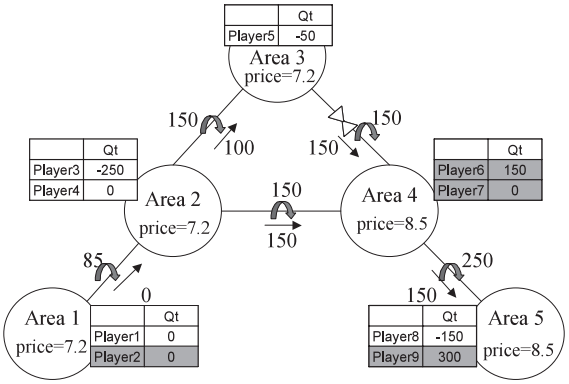


Fig. 14. Final solution

The final solution with the price for each area and the trade quantity for each player is described in Fig. 14.

As consequence on the congestion area 1, 2 and 3 have a low price of 7.2, all the players in those areas buy and sell electricity at the same price 7.2. On opposite area 4 and 5 see a higher price 8.5. Because of the congestion on the loop, player 6 is not able to buy all the quantity that he was asking, and player 1 is not selling electricity. The transmission capacity of lines 2-4 and 3-4 is utilized completely, other lines only partially. In particular, it may be pointed out how the congestion on line 1-2, which is present in the first procedure without any constraints is automatically relieved in the process of the split of the 2 submarkets.

In this example with few areas, only one split is enough to solve the market. However, in general, depending from the complexity of the system, it may happen that some lines are still congested in the submarkets created after the split. In these cases, it is necessary to apply the same procedure to the submarket and to divide it in other submarket until when all the transmission congestion are cleared.

6. Tests and Performaces

In order to describe the algorithmic process, a simple example is introduced in the previous section. In practice, the algorithm has to be utilized for larger systems with several areas with thousands of orders. A test system with 16 areas, 15 interconnection and 80,000 orders is developed. Using an Itanium 1 GHz CPU, the calculation CPU time is 0.01 sec. The result shows the high performance of the proposed algorithm compared with the previous results [(2)] where calculation time of the same order is necessary to solve markets with 9,000 orders. Considering that the CPU performance utilized in Ref. (2) is almost half of the performance of the CPU utilized in this paper it can be estimated that the proposed algorithm is 2-3 times faster than the method proposed in Ref. (2).

7. Conclusions

After an introduction of the auction rules for electric market, the congestion management is the one of the most important issues. In this paper, a new algorithm to identify the location at which the market has to be split duo to the transmission congestion is presented.

The algorithm was described not only for a radial network but also for a network with loops. In case of loops, a star-delta transformation is introduced to transform the network in a radial one. The method is simple, reproducible and always converges to a solution. The algorithm is tested on complex problems and the test results demonstrated the high performance of the algorithm.

The authors developed and implemented the method for the Japanese Electric Power Exchange that is in operation from April 2005.

The proposed algorithm is only applicable to radial networks or loops with 3 areas and at least one DC interconnection. This constraint fit with the Japanese interconnection network but can be a limitation in the implementation of a similar market in other countries. The authors are now investigating the possibility to solve mesh networks with the same methodology. In addition, the algorithm is implemented with the assumption of simple order, which is composed with one price and one quantity; it is possible to extend the algorithm to complex orders such as curve or multiple orders from the same player.

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References

(1) JEPX: “Trading operation guide”
(2) T. Sawa, et al.: “Market Splitting Single Auction in Wholesale Electric Power Exchange”, F408-A1 6-044 IEEJ General Meeting, Tokyo (2004-3) (in Japanese)
(3) T. Sawa, Y. Nakata, M. Tsurugai, and S. Sugiyama: “Fast Market Splitting Matching Method for Spot Electric Power Markrt”, *IEEJ Trans. PE*, Vol.127, No.4, pp.573–580 (2007-4) (in Japanese)
(4) S. Fukutome, et al.: “Bidding market based on single price model with network constraints”, PSCE 10-13 NY (2004-10)
(5) E. Wester: “Transmission pricing in Norway”, in Electricity Pricing and Technology, Kluwer Academic Publisher (1995)
(6) M. Friedman: A Program for Monetary Stability, Fordham University Press (1960)
(7) FERC SMD NOPR Docket, No.RM01-12-000
(8) ASX.: “Calculating opening and closing prices”, http://www.asx.com.au/investor/education/basics/open_Close.htm
(9) M. Tanimoto, et al.: “Japan Electric Power Exchange System”, *Mitsubishi Denki Giho*, Vol.80, No.3 (2006) (in Japanese)



Marta Marmiroli (Member) received her degree in Nuclear Engineering in 1996 from the University of Bologna, Italy. After a period of study at Tokyo University she joined Mitsubishi Electric Corporation in the 1997 working mainly on power market design and market supporting software. Her research interests include planning, power systems economics and energy risk.



Masahiko Tanimoto (Member) was born in Osaka, Japan in 1969. He received the B.S. and M.S. degrees in Electrical Engineering from Kyoto University, Japan in 1992 and 1994 respectively. He joined Mitsubishi Electric corp. in 1994 and is currently enrolled in Advanced Technology R&D center. He has been engaged in R&D on methodology, application software, and computer system for operation and control of power system including deregulation of electric power industry.

Yukitoki Tsukamoto (Member) received his M.S. in electric engineering form Waseda University, Tokyo, Japan in 1990. In the same year he joined Mitsubishi Electric Corporation as engineer. He was the project leader in the developmnt of the Japanese Electric Power Exchange system. He is currently manager of the system integration section. His research interests include energy economy, deregulation of the power system industry, assets management, and renewable energy. He is a member of CIGRE and IEE of Japan.



Ryuichi Yokoyama (Member) received the B.S., M.S. and Ph.D degrees in electrical engineering from Waseda University, Tokyo, Japan in 1968, 1970, and 1973, respectively. Since 1978 he has worked in the Faculty of Engineering of Tokyo Metropolitan University, as Professor of Electrical Engineering. From 2007 he is professor at Waseda University. His fields of interests include planning, operation control and simulation of large-scale power systems, and also analysis of deregulated electricity markets. He is a member of the Society of Instrument and Control Engineers (SICE) of Japan, the IEE of Japan, CIGRE, and IEEE.

